

NEUROSIM RESEARCH REPORT

INTELLIGENT EEG COGNITIVE ANALYTICS PLATFORM

Session ID:	VERIFY-001	Date:	2026-08-02
Duration:	05:00	Sampling Freq:	250 Hz
Signal Source:	Signal Simulator	Classification:	RELAXED

SPECTRAL BAND POWER BREAKDOWN

Band	Frequency Range	Relative Power (%)	Clinical Context
Delta (δ)	0.5 – 4.0 Hz	20.0%	Deep sleep, slow-wave activity
Theta (θ)	4.0 – 8.0 Hz	20.0%	Drowsiness, meditation, memory
Alpha (α)	8.0 – 13.0 Hz	45.0%	Relaxed alertness, calm focus
Beta (β)	13.0 – 30.0 Hz	15.0%	Active concentration, stress

AI NARRATIVE INTERPRETATION

The recorded synthetic signal showed predominantly alpha-band activity throughout the session, with moderate beta activity. The calculated feature profile remained stable, while the cognitive-load classifier remained within the moderate category.

RESEARCH SYSTEM DISCLAIMER: NeuroSim processes synthetic EEG-like signals for research, simulation, and analytical modeling. Analytical metrics do not constitute a medical diagnosis.